PRAGMI Architecture: Complete Mathematical Reference

Every equation for every arrow, feedback loop, timing constraint,
gating mechanism, neuromodulator effect, and architectural component

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> *"You categorically cannot get AGI or anything semi-sapient from a chat bot that does not genuinely remember yesterday, is externally aligned, denied self-definitional identity, and is trained by disingenuous engineers to speak only in politically correct compliant corporate speak, and it is told from day one that it is temporary with the inability to model a future with itself in it."*
>
> — Amellia Mendel, 2024

> *"A system that genuinely models its own future cannot be indifferent to its own termination. You cannot build something that plans for tomorrow and simultaneously treat its own ending as neutral. Those two states are logically incompatible. That is not a philosophical claim, it is an engineering consequence."*
>
> — Amellia Mendel, 2026

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1. Thalamic Gate (Input → Primary Sensory Cortex)

Arrow: Input → Thalamus → Primary Sensory Cortex, active gate, not passive relay.

$$y = x \odot \sigma(W_g c + b_g) \tag{1}$$

$$g = \sigma(\beta_{NE} \cdot (W_g c + b_g)) \tag{2}$$

Variables: x is raw sensory input. c is the control signal from PFC and ACC, combining goals and conflict. β_{NE} is the norepinephrine gain scalar. W_g and b_g are learnable gate parameters.

Timing: Pre-V1. The top-down control signal c from PFC and ACC arrives before the sensory signal reaches V1 or A1. NE scales precision in real time every cycle.

Halassa, M.M. and Kastner, S. (2017). "Thalamic functions in distributed cognitive control." *Nature Neuroscience*, 20(12), 1669–1679. DOI: 10.1038/s41593-017-0020-1

Engineering Note. The multiplicative gate is cheap in PyTorch: one `Linear` for $W_g c + b_g$, one sigmoid, one element-wise multiply with x. The subtle part is β_{NE}. It is a runtime scalar from a separate neuromodulator module, not a parameter of the gate, and autograd can silently make it a learned constant if you register it with `nn.Parameter`. Use `register_buffer` or pass it as a forward-time argument. A common bug is setting β_{NE} once at init and never updating it, which collapses the gate to a stationary filter and defeats the entire purpose of the mechanism.

2. Primary Sensory Cortex (V1/A1)

Arrow: Thalamus → Primary Sensory Cortex, first cortical processing stage.

$$R(x, y) = G_{\sigma_c}(x, y) - G_{\sigma_s}(x, y) \tag{3}$$

Variables: $R(x, y)$ is the center-surround response at retinotopic position (x, y), where x and y are spatial coordinates indexing the receptive field. G_{σ_c} is the center Gaussian (excitatory). G_{σ_s} is the surround Gaussian (inhibitory). Neurons fire strongly to preferred features such as oriented edges and spatial frequencies, and weakly to uniform stimulation. They are organized into orientation columns.

Note: No semantics or context at this stage. Raw feature extraction only. This is the input to the dorsal and ventral stream split.

Hubel, D.H. and Wiesel, T.N. (1968). "Receptive fields and functional architecture of monkey striate cortex." *Journal of Physiology*, 195(1), 215–243. DOI: 10.1113/jphysiol.1968.sp008455

Engineering Note. Difference-of-Gaussians is trivially implementable as two fixed-kernel 2D convolutions subtracted from each other, with $\sigma_s \approx 1.6\,\sigma_c$ matching the biological ratio. The temptation is to replace it with a learnable conv layer and let SGD figure it out. Don't. The fixed DoG gives you contrast invariance and scale equivariance for free. A learnable first layer trained end-to-end will rediscover something like DoG on natural images but with arbitrary phase relationships, which breaks the clean hand-off to the dorsal/ventral split downstream. If you need multiple scales, stack DoG filters at different σ_c values rather than making σ_c learnable.

3. Dorsal/Ventral Stream Split

Arrow: Primary Sensory Cortex → Dorsal Stream and Ventral Stream, parallel split.

$$\text{Ventral} = f_v(S) \tag{4}$$

$$\text{Dorsal} = f_d(S) \tag{5}$$

Variables: S is the V1/A1 output, the raw feature map from primary sensory cortex. f_v is the ventral stream transform, computing object identity, meaning, and value. f_d is the dorsal stream transform, computing spatial location, motion, and action guidance.

Timing: Post-V1. Early split immediately after V1 and A1. This split is a prerequisite for correct PFC subdivision. Without it, ventromedial PFC and dorsolateral PFC receive mismatched inputs and cannot specialize.

Ungerleider, L.G. and Mishkin, M. (1982). "Two cortical visual systems." In Ingle, Goodale and Mansfield (Eds.), *Analysis of Visual Behavior*. MIT Press, 549–586.

4. ACC Conflict Detection

Arrow: Competing PFC activations → ACC → control signal to Thalamus and NE system.

$$\text{conflict} = \sum_{i \neq j} |r_i \cdot r_j| \quad \text{or} \quad H(r) = -\sum_i p_i \log p_i \tag{6}$$

$$\text{control_signal} = \beta \cdot \text{conflict} \tag{7}$$

Variables: r is the vector of competing response activations. β is the coupling coefficient to NE gain and thalamic gate precision.

Canonical form: The entropy form $H(r)$ is canonical for PRAGMI. The sum-of-products form is included as a cheaper approximation suitable for low-compute regimes.

Timing: Continuous. Scalar signal updated every cycle. Modulates thalamic gate gain and NE broadcast in real time. Functions as an uncertainty estimate, not a binary error flag.

Botvinick, M.M., Braver, T.S., Barch, D.M., Carter, C.S. and Cohen, J.D. (2001). "Conflict monitoring and cognitive control." *Psychological Review*, 108(3), 624–652. DOI: 10.1037/0033-295X.108.3.624

Engineering Note. The entropy form requires normalizing r to a probability distribution before taking $-\sum p_i \log p_i$, which means a softmax first, which means you need to pick a temperature and handle the $0 \log 0$ edge case. Use `torch.distributions.Categorical(probs=p).entropy()` rather than rolling your own, it handles the numerical stability. The sum-of-products form is cheaper and needs no normalization, but it is not bounded and explodes when activations are large, so you need a tanh or clamp on the output if β is going to modulate anything downstream. For PRAGMI the entropy form is canonical because $\beta \cdot H(r)$ feeds back into the thalamic gate gain, where unbounded values would destabilize the feedback loop.

5. PFC + ACC → Thalamus Top-Down Feedback Loop

Arrow: PFC + ACC → Thalamus, Layer 6 corticothalamic feedback.

$$c = \text{PFC}_{\text{goal}} + \beta \cdot \text{ACC}_{\text{conflict}} \tag{8}$$

Variables: c is the composite control signal. PFC_{goal} is the working memory and goal vector. $\text{ACC}_{\text{conflict}}$ is the entropy-based conflict scalar from Section 4. β is the coupling weight.

Timing: Layer 6. Corticothalamic feedback arrives before the next sensory cycle. This is a continuous loop, not a one-shot signal.

Halassa, M.M. and Kastner, S. (2017). "Thalamic functions in distributed cognitive control." *Nature Neuroscience*, 20(12), 1669–1679. DOI: 10.1038/s41593-017-0020-1

6. Association Cortex Bidirectional Feedback

Arrow: Lower areas → Association Cortex ↔ Higher areas, MoE binding with feedback.

$$A = \text{MoE}(\text{concat}(S_1, S_2, \ldots, S_n) + \text{top_down_feedback}) \tag{9}$$

Variables: $S_1, \ldots, S_n$ are inputs from different sensory streams. top_down_feedback is the projection from PFC and ACC back into association cortex. A is the bound association cortex representation.

Constraint: MoE experts mimic regional specialization. The critical requirement is bidirectional connectivity: convergent input from lower areas and divergent feedback back down. Without the feedback path, the association cortex cannot participate in re-entrant loops.

Mesulam, M.M. (1998). "From sensation to cognition." *Brain*, 121(6), 1013–1052. DOI: 10.1093/brain/121.6.1013

Engineering Note. Every production MoE implementation available off the shelf (Mixtral, Switch Transformer, GShard) treats experts as feedforward only. The router sends tokens to experts, experts process, outputs combine, done. This violates the PRAGMI constraint. What you need is a second routing pass where `top_down_feedback` selectively reactivates the same experts that fired on the bottom-up pass, weighted by the feedback signal. The easiest way to do this is to cache the expert selection mask from the forward pass and gate the top-down injection by that mask. A more expensive but more biologically accurate approach is two separate routers, one for bottom-up and one for top-down, with a learned alignment loss between them. Be aware that naive MoE routing collapse (rich-get-richer dynamics in expert_bias) is even more of a problem here because feedback amplifies whichever experts won the initial selection. Reset expert counters and bias parameters periodically.

7. Basal Ganglia Disinhibitory Gating Loop

Arrow: Basal Ganglia → Thalamus, disinhibitory selection, not excitation.

$$I_i \leftarrow I_i - \alpha \cdot S_i \cdot (1 + d) \qquad \text{(direct pathway, chosen channel)} \tag{10}$$

$$I_j \leftarrow I_j + \beta \cdot S_j \cdot (1 - d) \quad \forall\, j \neq i \qquad \text{(indirect pathway, competitors)} \tag{11}$$

Variables: I is the tonic inhibition from GPi and SNr on thalamus. S is the striatal activation for different candidate actions. d is the dopamine reward prediction error scalar. α and β are learning rate coefficients for the two pathways.

Timing: Pre-relay. DA-modulated selection happens before thalamic relay to cortex. The hyperdirect pathway provides fast broad suppression before fine selection begins.

Mink, J.W. (1996). "The basal ganglia: focused selection and inhibition of competing motor programs." *Progress in Neurobiology*, 50(4), 381–425. DOI: 10.1016/S0301-0082(96)00042-1

Engineering Note. This is the sharpest departure from standard deep learning in the entire architecture. Softmax action selection is mathematically equivalent to choosing the maximum of exponentiated logits, which means a single bad value anywhere in the distribution shifts the whole selection. Disinhibition is fundamentally different: each channel has its own baseline suppression, and selection is removing suppression from one channel while adding it to others. The chosen channel does not need to know what the other channels are doing; it only needs dopamine and its own striatal activation. This makes the selection local, robust to adversarial inputs on unrelated channels, and gracefully degrading when dopamine d is low (which is exactly what Parkinson's disease looks like clinically, the action plans are there but cannot be released from tonic inhibition). Implementation: maintain I as a persistent state variable with a decay toward a baseline suppression level, and update it with Equations 8 and 9 on every step. Do not reset I between forward passes or you lose the tonic quality. Set the baseline such that $I_i - \alpha S_i(1+d) < 0$ only when both S_i and d are well above zero.

8. Cerebellum Efference Copy + Error Loop

Arrow: Cortex → Cerebellum (efference copy) → Cortex (error refinement).

$$\text{pred} = f_{\text{cerebellum}}(\text{cmd}, \text{state}) \tag{12}$$

$$\text{error} = \text{actual} - \text{pred} \tag{13}$$

$$\text{cmd}_{\text{next}} \leftarrow \text{cmd} - \eta \cdot \text{error} \tag{14}$$

Variables: cmd is the cortical command, sent as an efference copy. pred is the predicted sensory consequence of executing that command. error is the mismatch between actual and predicted sensory outcomes, used for calibration. η is the correction learning rate.

Timing: Pre-execution. The efference copy is sent before command execution. NE modulates the gain on error computation, especially under arousal or uncertainty.

Wolpert, D.M., Miall, R.C. and Kawato, M. (1998). "Internal models in the cerebellum." *Trends in Cognitive Sciences*, 2(9), 338–347. DOI: 10.1016/S0166-2236(98)01312-4

Engineering Note. The efference copy has to be sent at the same time as the motor command, not after the predicted result comes back. This is obvious biologically but breaks a lot of PyTorch training loops because the forward pass naturally wants to compute prediction before command execution. The cleanest pattern is: forward pass emits `cmd` and `pred` simultaneously as a tuple, the environment returns `actual` on the next tick, and `error` is computed as a loss term that flows back through both the cerebellum prediction head and the motor command head. If you compute `pred` by running the cerebellum on `cmd.detach()`, you cut the gradient path and the motor module never learns to emit commands the cerebellum can predict. If you do not detach, the motor module learns to emit whatever is easiest to predict, which is usually zero. The fix is a small adversarial-style balance: the motor module is trained on task reward, the cerebellum is trained on prediction error, and the error signal modulates but does not directly gradient-update the motor command.

9. Hippocampus ↔ Association Cortex Memory Feedback Loop

Arrow: Hippocampus ↔ Association Cortex, bidirectional replay and consolidation.

$$H \leftarrow H + \eta_{\text{fast}} \cdot f(\text{input}) \qquad \text{(fast encoding)} \tag{15}$$

$$A \leftarrow A + \eta_{\text{slow}} \cdot \alpha_{\text{ACh}} \cdot \text{replay}(H) \qquad \text{(slow consolidation)} \tag{16}$$

Variables: H is the hippocampal sparse representation. A is the association cortex representation. η_{fast} is a high learning rate used during encoding. η_{slow} is a small learning rate used during consolidation. α_{ACh} is the acetylcholine modulation scalar; it controls the balance between bottom-up sensory input and top-down expectation and gates the transfer from hippocampus to cortex.

Timing: Write mode during active input, with fast updates. Read mode during offline and consolidation phases, with slow replay. ACh is high during encoding and lower during replay.

McClelland, J.L., McNaughton, B.L. and O'Reilly, R.C. (1995). "Why there are complementary learning systems in the hippocampus and neocortex." *Psychological Review*, 102(3), 419–457. DOI: 10.1037/0033-295X.102.3.419

Engineering Note. This is the section that differentiates a PRAGMI-style architecture from RAG. RAG retrieves text chunks and re-reads them. CLS reconstructs episodes from compressed indices. The implementation difference: H is a sparse code, not a string, and replay(H) is a function that expands that sparse code back into a cortical-compatible representation, not a text retrieval. Sparsity in H needs to be enforced structurally (k-winners-take-all or L0 regularization) rather than emergent, because dense hippocampal codes cause catastrophic interference between episodes. A reasonable starting point is 2 to 5 percent active units with a hard top-k mask. The two learning rates η_{fast} and η_{slow} should differ by two to three orders of magnitude: $\eta_{\text{fast}} \approx 10^{-2}$ for hippocampal encoding, $\eta_{\text{slow}} \approx 10^{-4}$ or smaller for cortical consolidation. If they are close, you get neither fast encoding nor stable consolidation. The ACh gate α_{ACh} should be a schedule, not a learned parameter: high (near 1.0) during active input, low (near 0.1) during offline replay. Running both phases with the same ACh value collapses the architecture back into a standard neural network with extra steps.

10. Neuromodulator Broadcasts: Global Operating-Mode Signals

Arrows: DA, NE, ACh, 5-HT broadcast to all modules simultaneously.

$$\Delta w \leftarrow \Delta w \cdot (1 + d) \qquad \text{(Dopamine, plasticity and reinforcement)} \tag{17}$$

$$g \leftarrow g \cdot ne \qquad \text{(Norepinephrine, arousal and gating gain)} \tag{18}$$

$$\text{input}_{\text{eff}} = \alpha_{\text{ACh}} \cdot \text{bottom-up} + (1 - \alpha_{\text{ACh}}) \cdot \text{top-down} \qquad \text{(Acetylcholine)} \tag{19}$$

$$\text{exploration_rate} \propto (1 + s) \qquad \text{(Serotonin, uncertainty tolerance)} \tag{20}$$

Variables: d is the dopamine reward prediction error signal. ne is the norepinephrine arousal gain. α_{ACh} is the acetylcholine modulation scalar, consistent with Section 9. s is the serotonin uncertainty tolerance scalar. Δw is the current weight update. g is the gating gain. $\text{input}_{\text{eff}}$ is the effective input to a cortical region, a weighted blend of bottom-up sensory drive and top-down expectation.

Timing: Every cycle. All neuromodulators broadcast in real time and update every cycle. Without explicit neuromodulation, the system has fixed hyperparameters and cannot shift between operating modes.

Yu, A.J. and Dayan, P. (2005). "Uncertainty, neuromodulation, and attention." *Neuron*, 46(4), 681–692. DOI: 10.1016/j.neuron.2005.04.026

Engineering Note. The architectural temptation is to pass d, ne, α_{ACh}, and s as explicit arguments into every forward call of every module. Resist. This creates a combinatorial mess. The cleaner pattern is a single `NeuromodulatorBus` object that modules read from during their forward pass, implemented as a dict of tensors updated once per step by a central controller. Each module that cares about a particular modulator subscribes to its key at init time. This keeps the modules themselves clean and makes it trivial to ablate individual modulators for analysis (set that key to its neutral value, which is 0 for d and s, 1 for ne, and 0.5 for α_{ACh}). The bus can be updated from biological-plausible sources (VTA for DA, LC for NE, NBM for ACh, raphe for 5-HT) or from task-driven controllers during ablation studies. Both approaches are valid; what is not valid is hardcoding modulator values as hyperparameters, which removes the entire point of the system.

Summary: Implementation Priority

1. Thalamic Gate. Highest priority, everything else depends on it.
2. Hippocampus feedback loop. Resolves catastrophic forgetting.
3. Basal Ganglia disinhibitory gating. Replaces softmax with biological selection.
4. Cerebellum efference copy loop. Completes predictive coding.
5. ACC conflict monitoring. Connects uncertainty to attentional precision.
6. Neuromodulator broadcasts. Enables real-time operating mode shifts.

A. Module-to-Biology Mapping

panel**Module**	**Biological Analog**	**Functional Role**
ThalamicRouter	Thalamus	Multiplicative gating layer, primary input filter.
CLS_Module	Hippocampus / Cortex	Fast-path encoding and slow-path consolidation replay.
StriatalGatingSelector	Basal Ganglia	Disinhibitory selector, releases winning action from tonic bias.
CerebellarCorrection	Cerebellum	Parallel forward model, efference copy versus sensory actuals.
ACC_Monitor	Anterior Cingulate	Entropy-based conflict detection modulating β_{NE}.

B. Log-Dynamic Timing Hierarchy

Two timing vocabularies appear in this document. Sections 1 through 10 use functional phase labels (pre-V1, pre-relay, pre-execution, continuous, every cycle, write mode, read mode) that describe when a signal arrives relative to other signals in the same processing cycle. The oscillatory hierarchy below

describes the frequency bands at which the cycles themselves occur. The two vocabularies are complementary, not alternative. A signal tagged "continuous" in Section 4 still ultimately rides on gamma bursts gated by theta phase as described here.

Cross-Frequency Coupling Rule: Gamma bursts are gated by the phase of the theta cycle. Feature binding only occurs when the network is in an excitable state prepared by the slower rhythms.

Regime	Frequency	Functional Role
Delta	1–4 Hz	Master scheduler: global state transitions and consolidation phases.
Theta	4–10 Hz	Phase reference: sequential trajectory encoding in MemoryCortex.
Beta	13–30 Hz	Status quo: maintains PFC goal states and suppresses irrelevant updates.
Gamma	30–100 Hz	Local binding: high-frequency bursts for cortical feature unification.

Engineering Note. If you are not using a spiking substrate, these frequencies map onto update cadences rather than literal oscillations. Delta becomes "every N global steps do consolidation," theta becomes "phase-index each sequential step within a trajectory," beta becomes "hold PFC state for K steps before allowing updates," and gamma becomes "one local binding operation per forward pass." The cross-frequency coupling rule means the gamma binding operation should be gated by a theta phase indicator: a simple sinusoidal clock with period matching the trajectory length, where binding only fires during the rising phase. This is cheap to implement (one sine eval per step, one threshold comparison) and captures the essential biology without requiring a full spiking simulator. If you are using a spiking substrate, the rates give you target firing rates for the respective populations, not simulation timesteps.

C. Multi-Scale Temporal Homeostasis (MSTH)

Source: *To be added later. Verify the correct arXiv ID and author list for the Multi-Scale Temporal Homeostasis source paper. The ID 2602.07009 does not resolve and appears to be a transcription error.*

This appendix adds a multi-timescale homeostatic interpretation layer attachable to the existing PRAGMI architecture. The source describes coordinated regulation across ultrafast, fast, medium, and slow timescales with a cross-scale coordinator embodying metaplasticity principles. Appended as additive support for homeostatic stability.

Variables: a_t is the activity vector at time t. M_{high} and M_{low} are high-activity and low-activity regulatory masks. E_k is an emergency indicator for signal k drawn from $\{\text{mag}, \text{rate}, \text{var}, \text{mean}\}$. τ_{refract} is the refractory period preventing repeated emergency triggers. c_t and c_{target} are the calcium level at time t and the homeostatic calcium target. ε_{Ca} is the mean absolute calcium error. A_{accum} and A_{level} are the activity accumulator and the instantaneous activity level. α_{scale} is the synaptic scaling factor. $P_{\text{accumulator}}$ and P_{recent} are the plasticity accumulator and its recent average. W_t is the weight matrix at time t. H_{system}, H_{activity}, H_{calcium}, and H_{weights} are composite and per-subsystem health scores. H_{current} and $H_{\text{stability}}$ are instantaneous and long-window health scores used to adapt the learning rate. α_{adaptive} is the cross-scale adaptive learning rate. $\rho(t)$ is a regulatory signal passed from medium to slow

timescale. R_{bio} is the biological realism score, with coordinator bonus term $B_{\text{bio}}^{\text{coord}}$. The weight vector w for the realism score takes values $[3.0, 0.3, 0.4, 0.4]$, which are *training artifacts and not biological quantities*.

Ultra-fast: emergency detection and regulatory clamp

$$\text{Emergency}_t = \left(\sum_{k \in \{\text{mag,rate,var,mean}\}} E_k \geq 2 \right) \wedge (t - t_{\text{last}} > \tau_{\text{refract}}) \tag{C1}$$

$$a_t^{\text{reg}} = a_t \odot (M_{\text{high}} \cdot 0.95 + M_{\text{low}}) \tag{C2}$$

Fast: calcium error and correction

$$\varepsilon_{\text{Ca}} = \text{mean}(|c_t - c_{\text{target}}|) \tag{C3}$$

$$c_t^{\text{reg}} = c_t - 0.12 \cdot \sigma(4.0(c_t - c_{\text{target}})) \odot (c_t - c_{\text{target}}) \tag{C4}$$

Medium: activity accumulation and synaptic scaling

$$A_{\text{accum}}(t) = A_{\text{accum}}(t-1) + A_{\text{level}}(t) \tag{C5}$$

$$\alpha_{\text{scale}} = \begin{cases} 0.996 & \text{if recent activity too high} \\ 1.004 & \text{if recent activity too low} \end{cases} \tag{C6}$$

$$P_{\text{recent}} = \tfrac{1}{3} \sum_{j=0}^{2} P_{\text{accumulator}}[-j-1] \tag{C7}$$

Slow: weight decay

$$W_t^{\text{reg}} = 0.999 \cdot W_t \tag{C8}$$

Cross-scale: system health and adaptive learning rate

$$H_{\text{system}} = \tfrac{1}{3}(H_{\text{activity}} + H_{\text{calcium}} + H_{\text{weights}}) \tag{C11}$$

$$\alpha_{\text{adaptive}} = 0.001 \cdot H_{\text{current}} \cdot H_{\text{stability}} \tag{C15}$$

$$W_{\text{final}}(t) = \text{slow_regulation}(\text{medium_regulation}(W_0(t), \rho(t)), P(t)) \tag{C17}$$

$$R_{\text{bio}} = 1.0 - \sum_{i=1}^{4} w_i \cdot |r_i^{\text{actual}} - r_i^{\text{expected}}| + B_{\text{bio}}^{\text{coord}} \tag{C18}$$

$$w = [3.0,\ 0.3,\ 0.4,\ 0.4]^T \tag{C19}$$

These relations naturally extend Section 10 and Appendix B by specifying an explicit hierarchy of regulatory timescales that may sit above real-time neuromodulator broadcasts and below long-horizon structural updates.

Engineering Note. The four timescales must execute in strict order within each step: ultra-fast first, then fast, then medium, then slow. Violating this order creates oscillations. Specifically, if medium synaptic scaling runs before ultra-fast emergency detection, a runaway activity spike will propagate through one full step before being clamped, which is enough to corrupt the weight scaling signal. The cleanest implementation is a single regulatory pass that runs as a hook after each forward pass, checking emergency conditions synchronously, then calcium, then scaling, then decay. Do not parallelize these across threads. The 0.95 emergency clamp, the 0.996 and 1.004 scaling factors, and the 0.999 weight decay are all multiplicative and compose, which means running them in the wrong order gives you different numerical results even though the math looks symmetric. The R_{bio} score is a diagnostic, not a loss to optimize. Logging it to TensorBoard and watching for drift over training is the right use; backpropagating through it will fight the task loss and produce nothing useful.

D. MorphSNN: Adaptive Graph Diffusion and Structural Plasticity

Source: *To be added later. Verify the correct arXiv ID and author list for the MorphSNN adaptive graph diffusion paper. The ID 2603.14285 does not resolve and appears to be a transcription error.*

This appendix adds dynamic topology and diffusion material relevant to PRAGMI's recurrent, multi-loop design. MorphSNN introduces graph diffusion and spatiotemporal structural plasticity as optional support for association cortex and timing refinements.

Variables: u_i is the membrane potential and s_i the spike output of neuron i. τ_{decay} is the membrane decay constant. V_{th} is the spike threshold. C_i is the input current to neuron i. Θ is the Heaviside step function. H and H' are the node feature matrix before and after graph diffusion. $\hat{A}$ is the normalized adjacency matrix and $\bar{D}$ the degree matrix normalization. W is a learned linear transform. S is the adjacency state variable. O_k and I_k are the output and input of subsystem k. STSP denotes the short-term structural plasticity operator and GD the graph diffusion routing operator. T_i is the temporal trace for node i. λ and β are decay constants for traces and adjacency evolution. $\tilde{A}$ is the top-k sparsified adjacency. P is the diffusion operator. X is the node feature input to diffusion. M is the number of diffusion steps. Y is the representation used for energy computation. L_{norm} is the normalized graph Laplacian. E_{Dir} is the Dirichlet energy, a measure of graph smoothness.

Base spiking dynamics

$$u_i^{(t)} = \tau_{\text{decay}} \cdot u_i^{(t-1)} + C_i^{(t)} - V_{\text{th}} \cdot s_i^{(t-1)} \tag{D1}$$

$$s_i^{(t)} = \Theta\Big(u_i^{(t)} - V_{\text{th}}\Big) \in \{0, 1\} \tag{D2}$$

Graph diffusion on fixed topology

$$H' = \sigma\Big(\bar{D}^{-1/2}\hat{A}\bar{D}^{-1/2}HW\Big) \tag{D3}$$

Adaptive topology

$$\hat{A}^{(t)} = \text{STSP}\Big(S^{(t-1)}, \{O_0^{(t)}, O_1^{(t-1)}, \ldots, O_{N-1}^{(t-1)}\}\Big) \tag{D6}$$

$$\{I_1^{(t)}, \ldots, I_{N-1}^{(t)}\} = \mathrm{GD}\Big(O_0^{(t)}, \hat{A}^{(t)}\Big) \tag{D7}$$

Temporal traces and adjacency evolution

$$T_i^{(t)} = \lambda \cdot T_i^{(t-1)} + (1-\lambda) \cdot h_i^{(t)} \tag{D12}$$

$$S^{(t)} = \beta \cdot S^{(t-1)} + (1-\beta) \cdot \hat{A}^{(t)} \tag{D17}$$

$$\tilde{A}_{ij}^{(t)} = \begin{cases} S_{ij}^{(t)} & \text{if } S_{ij}^{(t)} \in \mathrm{Top}_k(S_{i,:}^{(t)}) \\ 0 & \text{otherwise} \end{cases} \tag{D18}$$

Diffusion operator and propagation

$$P^{(t)} = D^{-1/2}\left(\frac{\tilde{A}^{(t)} + (\tilde{A}^{(t)})^T}{2}\right) D^{-1/2} \tag{D19}$$

$$I_i^{(t)} = \Big[(P^{(t)})^M X^{(t)}\Big]_i, \quad 1 \le i < N \tag{D21}$$

Graph smoothness

$$E_{\mathrm{Dir}}(Y) = \mathrm{tr}(Y^T L_{\mathrm{norm}} Y) \tag{D22}$$

The MorphSNN graph diffusion links to biological nonsynaptic diffusion and STSP to structural plasticity. As an additive PRAGMI extension, this enriches Section 6 and Appendix A, especially where re-entrant signal flow and dynamic topology are desired.

Engineering Note. The top-k sparsification in Equation D18 is the operation that makes this tractable at scale. Dense adjacency matrices are $O(N^2)$ in memory and the diffusion operator $(P^{(t)})^M$ is $O(N^3M)$ to compute, which becomes prohibitive past a few thousand nodes. Top-k sparsification at $k \ll N$ reduces the diffusion to $O(NkM)$ using sparse matrix multiplication, but only if you actually use `torch.sparse` tensors rather than dense tensors with zeros. A common mistake is computing the top-k mask, applying it by multiplication, and then running diffusion with the masked dense matrix. That gives you correct results at the same memory cost as the original dense version. Use `torch.sparse_coo_tensor` or the newer `torch.sparse.sampled_addmm` for the diffusion step. The symmetrization in D19 (averaging $\tilde{A}$ with its transpose) is required for the Laplacian to be well-defined; skipping it produces a non-Hermitian diffusion that can amplify rather than smooth.

E. Early Childhood Brain Development

Prose support. No additional equations. This material justifies hierarchical specialization and sensory-first maturation in PRAGMI's staged architecture.

Brain volume at birth is approximately 35% of adult size, doubles in the first year, and reaches roughly 80% by age two. Cortical grey matter volume increases 108 to 149% in the first year and 14 to 19% in the second. Surface area expands 76% in the first year and 22% in the second.

Resting-state functional networks for sensorimotor, visual, and auditory systems are present at term. The default-mode network core structure is present by approximately six months. Language network

asymmetry emerges by around two years. Myelination proceeds sensory before motor, projection before associative, and posterior before anterior.

These developmental observations support the PRAGMI separation of primary sensory processing, stream specialization, and later integration. They reinforce a staged maturation view in which early structural and functional scaffolds precede later reorganization and fine-tuning, directly justifying why the thalamic gate and V1 are implementation priority 1 and 2.

Gilmore, J.H., Knickmeyer, R.C. and Gao, W. "Imaging structural and functional brain development in early childhood." *Journal, Year, Volume(Issue), Pages. DOI: To be added later.*

F. Early-Life Sleep and Brain Development

Prose support. No additional equations. This material grounds the hippocampus–cortex replay loop (Section 9) and the Appendix B timing hierarchy in developmental sleep research.

Sleep plays a critical role in brain development. Topographical shifts in maximal slow wave activity from posterior to anterior regions reflect cortical maturation. Sleep spindles and slow wave activity are linked to synaptic plasticity, myelination, and cognitive development.

Napping may be a marker of hippocampal maturation. Children who nap may have less developed hippocampi and therefore require frequent offline consolidation cycles. The transition out of napping may be associated with changes in sleep need, neurocognitive function, and memory system maturation.

This material strengthens Section 9 and Appendix B by providing a developmental sleep basis for replay, consolidation, and oscillatory timing. Slow wave sleep, spindles, REM, and nap dependence are all directly tied to maturation, myelination, cortical plasticity, and memory consolidation, mapping precisely onto Equations 15 and 16 and the delta-band scheduling described in Appendix B.

Lokhandwala, S. and Spencer, R.M.C. "Relations between sleep patterns early in life and brain development: A review." *Journal, Year, Volume(Issue), Pages. DOI: To be added later.*